

HALCYON FLIGHTWORKS

A320neo

Flight Crew Operating Manual



HALCYON A320NEO

Contents

1	General
2	Aircraft Systems
3	Normal Procedures
4	Performance
5	Limitations

This manual describes the Halcyon A320neo as it is simulated. It follows the order a crew meets the aircraft: what it is, how its systems behave, how it is flown on a normal day, what it can lift and where its limits are.

Values are rounded for readability. Where the simulation departs from the published type data, the text says so.

NOTE Every procedure in chapter 3 is also available as an interactive checklist on the EFB.

1 General

Introduction

The A320neo is a twin-engine, single-aisle airliner with fly-by-wire controls and a two-crew cockpit. The Halcyon model simulates the CFM LEAP-1A variant with sharklets, a two-class cabin for 174 passengers and the full set of normal and abnormal procedures a line crew uses.

The flight deck is modelled for both seats. Every switch, knob and guarded button is clickable, and the overhead panel follows the dark cockpit philosophy: in normal operation no light is on.

Principal dimensions

Item	Value
Overall length	37.57 m
Wingspan (with sharklets)	35.80 m
Height	11.76 m
Wheel track	7.59 m
Cabin width	3.70 m
Turning radius (wingtip)	20.4 m

Cockpit layout

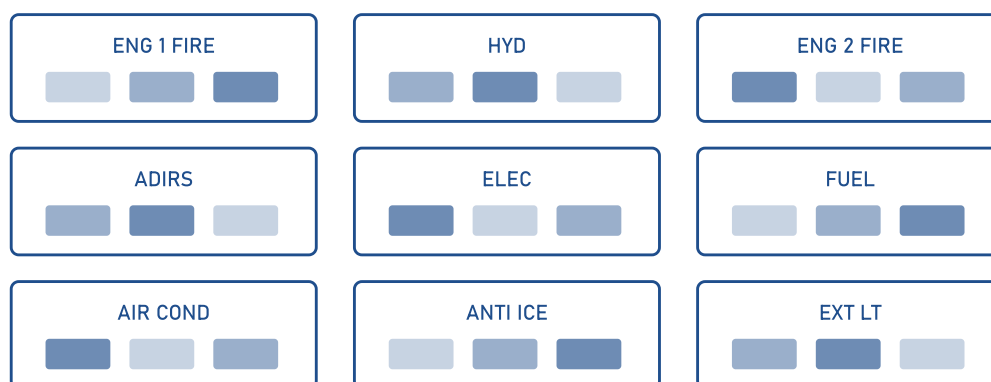


Figure 1-1 · Overhead panel, simplified

The overhead panel groups the systems by area, fore to aft: engine fire, hydraulics and electrics in the centre, air conditioning and anti-ice aft. Lights on a pushbutton mean the crew must act; a white OFF means a system was deliberately switched off.

How to read this manual

Chapters open with a short description and then list the items in the order they are checked. Items in capitals are the labels printed on the panel. Notes add information; cautions protect the aircraft; warnings protect the people on board.

CAUTION The simulated engines reach operating temperature in about two minutes. Applying takeoff thrust earlier shortens their simulated life and raises a maintenance message on the EFB.

2 Aircraft Systems

Electrical

Two engine-driven generators and the APU generator feed two independent AC networks. The batteries keep the essential buses alive while the aircraft is cold and dark.

Component	Value
GEN 1, GEN 2	90 kVA each
APU GEN	90 kVA
BAT 1, BAT 2	23 Ah, 25.5 V minimum
Static inverter	Essential AC on batteries only

The simulation models the normal and failed states of every component in the table. Failures can be triggered from the EFB failures page or left to occur at random with the wear setting.

Hydraulic

Green, blue and yellow systems at 3,000 psi. The power transfer unit keeps green and yellow pressure balanced when one engine is shut down.

Component	Value
Green	Engine 1 pump, PTU
Blue	Electric pump, RAT
Yellow	Engine 2 pump, electric pump
PTU	Automatic above 500 psi difference

The simulation models the normal and failed states of every component in the table. Failures can be triggered from the EFB failures page or left to occur at random with the wear setting.

Fuel

Two wing tanks and a centre tank. The centre tank feeds first while the wing pumps stay on to keep the engines supplied if the centre pumps fail.

Component	Value
Wing tanks	13,700 kg total
Centre tank	6,500 kg
Crossfeed	Manual, one valve
Unusable fuel	About 60 kg

The simulation models the normal and failed states of every component in the table. Failures can be triggered from the EFB failures page or left to occur at random with the wear setting.

Air conditioning

Two packs cooled by ram air supply three zones. Hot air is trimmed per zone from the cockpit panel.

Component	Value
Packs	2, automatic flow control
Zones	Cockpit, forward cabin, aft cabin
Temperature range	18 to 30 °C

The simulation models the normal and failed states of every component in the table. Failures can be triggered from the EFB failures page or left to occur at random with the wear setting.

Autoflight

Two flight guidance computers drive the autopilot and autothrust. The FCU sets targets; the flight mode annunciator shows what the aircraft is really doing.

Component	Value
AP 1, AP 2	Either one engaged, both for autoland
A/THR	Active between idle and CL detent
Modes	SPD, HDG, ALT, V/S, FPA, LOC, G/S, LAND

The simulation models the normal and failed states of every component in the table. Failures can be triggered from the EFB failures page or left to occur at random with the wear setting.

3 Normal Procedures

Normal procedures are flown from memory and confirmed with the checklist. The pilot flying calls, the pilot monitoring responds and moves the control. The flows below are grouped by phase.

COCKPIT PREPARATION	
BAT 1, BAT 2	AUTO
EXT PWR	ON
ADIRS 1, 2, 3	NAV
NAV and LOGO lights	ON
Fuel quantity	CHECK
Oxygen crew supply	ON
FMS	INITIALISED
Takeoff data	SET AND CROSS-CHECKED

BEFORE START	
Cockpit preparation	COMPLETED
Seat belts sign	ON
Doors	CLOSED
Beacon	ON
Thrust levers	IDLE
Parking brake	AS REQUIRED
Transponder	AUTO

NOTE The APU bleed can stay on for engine start. Switch it off after both engines are stable to save fuel.

Engine start

Engine 2 is started first so the yellow system powers the brakes while the ground crew is still near the nose. The start is automatic: the FADEC opens the start valve, adds fuel at the right N2 and aborts on its own if the temperature or the rotation is wrong.

ENGINE START	
ENG MODE selector	IGN/START
ENG 2 master switch	ON
N2, EGT, fuel flow	MONITOR
ENG 1 master switch	ON

ENG MODE selector	NORM
APU bleed	OFF

Taxi and takeoff

Taxi on the nose-wheel tiller with a maximum of 30 knots in a straight line and 10 knots in turns. Before entering the runway, check the takeoff configuration with the T.O CONFIG button.

BEFORE TAKEOFF	
Flight controls	CHECKED
Flaps	SET
T.O CONFIG	TEST
Cabin	READY
Strobe lights	ON
TCAS	TA/RA

4 Performance

The tables give approximate takeoff speeds for a dry runway at sea level with CONF 1+F and TOGA thrust. The EFB performance page computes exact values for the day and should be preferred.

Takeoff mass (t)	V1 (kt)	VR (kt)	V2 (kt)	Field length (m)
56	126	127	132	1560
60	131	132	137	1690
64	136	137	142	1830
68	141	142	146	1990
72	146	147	151	2170
76	151	152	156	2370
79	154	155	159	2520

Landing

Landing mass (t)	VLS CONF FULL (kt)	VAPP (kt)	Landing distance (m)
52	118	123	1180
56	122	127	1240
60	126	131	1310
64	130	135	1390
67	133	138	1450

CAUTION Add 5 knots to VAPP in gusty wind, and never less than a third of the steady headwind.

5 Limitations

Limit	Value
Maximum takeoff mass	79,000 kg
Maximum landing mass	67,400 kg
Maximum zero fuel mass	64,300 kg
VMO / MMO	350 kt / M 0.82
Maximum operating altitude	39,800 ft
Maximum crosswind, takeoff and landing	38 kt
Maximum tailwind	15 kt
Runway slope	±2 %
Maximum taxi speed in a turn	10 kt

WARNING Exceeding the limits above in the simulation triggers the corresponding structural or engine failure when wear is enabled.